

Alejandro Vargas

Software Engineer · .NET, React & Applied AI

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PROFILE

Computer Systems Engineer with more than 5 years of experience in software development, focused on building web applications, APIs, and solutions driven by data and artificial intelligence.

Experienced with .NET, React, Python, Rust, and SQL, as well as cloud and development tools such as Azure, Docker, and Git. Focused on designing scalable, maintainable solutions centered on solving real business needs.

EXPERIENCE

Programmer · Avance Enterprise Solutions (AVNC) 12/2025 – Present

- Design and development of a CRM system.

Web & Tools Developer · Ferrexpress 12/2023 – 12/2025

- Led a development team building web applications with .NET and Blazor, implementing Azure services.
- Implemented scalable, high-performance solutions with ASP.NET Core, React, and SQL databases.

PUBLICATIONS

Intelligent Agents for Mental Health: Depression Detection Through ML and LLM Model Fusion in Audio Data

ICITS 2026 · Springer · LNNS 1967 · Roatán, Bay Islands, Honduras · 2026 · **Published**
doi.org/10.1007/978-3-032-27541-7_26

Intelligent Optimization of Garbage Collection Routes through RAG Architecture in San Pedro Sula (Smart City)

IEEE LA-CCI 2025 · IEEE · Mexico City, Mexico · 2025 · **Published**
doi.org/10.1109/LA-CCI66231.2025.11270447

Early Prediction of ADHD Using Machine Learning Techniques in Educational Contexts with Limited Resources: Case of San Pedro Sula, Honduras

CISTI 2025 · Springer · LNNS 1717 · Lisbon, Portugal · 2025 · **Published**
doi.org/10.1007/978-3-032-10721-3_14

Spatio-Temporal DeepONets with Graph Neural Networks for Probabilistic Dengue Forecasting in Honduras

Maastricht, Netherlands · 2026 · **Pending publication**

Agentic Detection of Progressive Radicalization in User Posting Trajectories: A Hybrid LLM and Rule-Based Approach

Lima, Peru · 2026 · **Pending publication**

FEATURED PROJECTS

Ferrexpress Sales Website

2024 · In production

A web platform with an interactive catalog, a shopping cart with online payments through Stripe, and order-tracking and back-office tools, built with Blazor and ASP.NET Core on SQL Server in Azure.

Blazor · ASP.NET Core · SQL Server · Azure · Stripe API

SKILLS

Languages & Frameworks

C#, .NET, ASP.NET Core, Blazor, Rust, React, Next.js, React Native, JavaScript, TypeScript, Python, Flutter

Artificial Intelligence

Machine Learning, LLMs, RAG, Agents

Databases

SQL Server, PostgreSQL, Azure SQL, Firebase, Entity Framework Core

Cloud & DevOps

Azure, Docker, Git, GitHub, CI/CD

Architecture & Methodologies

REST APIs, OOP, SOLID, Scrum

EDUCATION

Computer Systems Engineering

Universidad Jesús de Nazareth
2025

High School Diploma in Sciences and Humanities

Angels Bilingual School
2019–2021

CERTIFICATIONS

International Course on Information Security, Cybersecurity and Privacy Protection

Interpretation of the ISO/IEC 27001:2022 standard
2024

LANGUAGES

Spanish · Native
English

RESEARCH AREAS

Software Engineering, Machine Learning, Data Science, System Architecture, Cloud Computing, Web Development

ASEGURE Payroll Automation

2023 · Active

C# tools with batch processing, automated calculations, and optimized SQL queries, integrated with the ERP and with Power BI reporting.

C# · SQL Server · Entity Framework · Azure SQL · Power BI

References available upon request

Ferrexpress Freight Management Web App

2024 · In production

A web app to create and control freight orders, manage routes, and assign drivers and vehicles, with real-time delivery tracking through SignalR and Google Maps.

Blazor · ASP.NET Core · SQL Server · Azure · SignalR · Google Maps API

Ferrexpress Construction Projects ERP

2026 · Delivered

A Blazor (.NET 10) web ERP on PostgreSQL: projects with a status workflow and stage-based progress, versioned budgets, finances with approvals and budget alerts, purchase orders with partial receipts, a site log, and versioned documents, with role-based access.

Blazor · .NET 10 · ASP.NET Core Identity · PostgreSQL · EF Core · Azure Blob Storage

Trankipay

2026

A bilingual site where patients browse, compare, and buy packages; prices are computed on the server (package or age-based discounts, plus sales tax), and an admin panel manages packages, orders, and users by role.

React · Vite · Tailwind CSS · Node.js · Express · SQL Server · JWT · Google OAuth

Offline-First Field Data App for EUDR

2026 · Field pilot

An offline-first Flutter mobile app — conditional survey, GPS-walked plot boundary, and consent capture — with encrypted storage and a queue that syncs to a FastAPI/PostgreSQL backend without duplicating records.

Flutter · Dart · Hive · Python · FastAPI · PostgreSQL